Simulating Random Graph Models in Python

Dissertation Presentation

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Why Study Random Graph Models?

Networks are everywhere:

- Social networks, the internet, neural networks, citation graphs
- All share structural properties that deterministic models cannot capture

Key structural questions:

- How do large networks form and evolve?
- Why do real-world networks have short average path lengths (small-world property)?
- Why do a small number of nodes attract most connections (hubs)?

Erdős–Rényi
Random edge probability

Watts–Strogatz
Small-world rewiring

Barabási–Albert
Preferential attachment

Goal: Understand the mathematical foundations of three principal random graph models, test theoretical predictions through Python simulation with hundreds of trials per parameter value, and validate against known results.

The Erdős–Rényi Model $G(n, p)$

Definition: Vertices are connected independently with probability p.

Key Metrics:

- Expected Edges: $p\binom{n}{2}$
- Avg. Degree: $p(n-1)$
- Threshold: $p_c = \ln(n)/n$

The Poisson Limit:
In sparse graphs ($p = \lambda/n$), degrees follow a Poisson distribution. *Result: No heavy tails or hubs.*

Erdős–Rényi: Phase Transitions

Subcritical ($\lambda < 1$)

- Small components
- Sizes: $O(\log n)$

Critical ($\lambda = 1$)

- Giant emerges
- Size: $O(n^{2/3})$

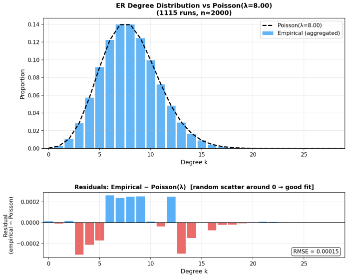
Supercritical ($\lambda > 1$)

- Unique giant
- Linear size: $O(n)$

Real-World Gap: $G(n, p)$ misses the **high clustering** and **heavy tails** seen in actual social or web networks.

ER Simulation: Degree Distribution Confirms Poisson

1115 runs · $n = 2000$ · $\lambda = 8.00$ · $p = \lambda/n = 0.004$



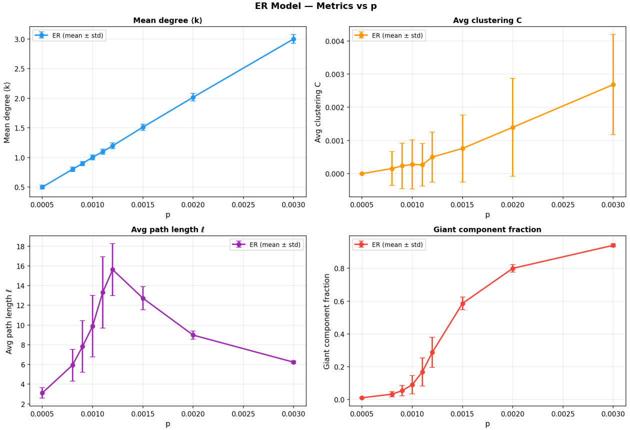
1115
simulation runs

RMSE
$= 0.00015$

Near-perfect
theory fit

Conclusion: Empirical degree distribution matches $\mathrm{Poisson}(8.00)$ almost exactly. Residuals show random scatter around zero which confirms the Poisson limit theorem numerically.

ER Phase Transition: Giant Component Emergence (RQ1)



RQ1 Answer: empirical 0.5 threshold at $p \approx 0.0014$ vs theory $p_c = 1/n = 0.001$; the difference is consistent with finite-size effects at $n = 1000$.

The Watts–Strogatz Small-World Model

Construction: (1) Begin with a regular ring lattice with n vertices, each connected to k nearest neighbours. (2) Rewire each edge with probability β to a uniformly chosen random vertex (no self-loops or duplicates).

Effect of rewiring parameter β:

$\beta = 0$

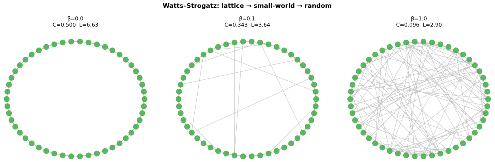
Pure ring lattice: HIGH clustering, LARGE average path length.

$0 < \beta \ll 1$

Small-world regime: a few long-range shortcuts reduce L dramatically while C stays high.

$\beta = 1$

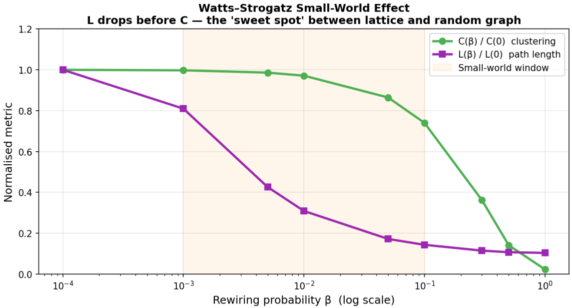
Effectively random: LOW clustering, SHORT path length.



$\beta = 0$: pure lattice ($C = 0.5$, $L = 6.63$) $\rightarrow$ $\beta = 0.1$: small-world ($C = 0.343$, $L = 3.64$) $\rightarrow$ $\beta = 1$: random ($C = 0.096$, $L = 2.90$)

WS Simulation: The Small-World Window (RQ2)

$n = 500 \cdot k = 8 \cdot \beta$ swept log-uniformly from 10^{-4} to $1 \cdot 30$ runs per point



RQ2 Answer

Small-world window:

$10^{-3} < \beta < 10^{-1}$

β	C/C_0	L/L_0
10^{-3}	1.00	0.81
10^{-1}	0.75	0.14

At $\beta = 10^{-3}$: L already falling, C fully intact.

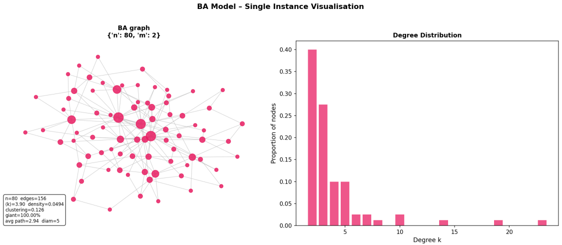
At $\beta = 10^{-1}$: L at 14%, C still 75%.

Key insight: L drops sharply near $\beta \approx 10^{-3}$ because even a few shortcuts bridge distant regions, while C stays high as local lattice structure is preserved.

The Barabási–Albert Preferential Attachment Model

Construction: (1) Begin with a small initial connected graph on m_0 vertices. (2) At each step, add one new vertex with $m \leq m_0$ edges. (3) Each new edge connects to vertex i with probability $\Pi(i) = d_i \big/ \sum_j d_j$.

Intuition: "The rich get richer." High-degree vertices attract more connections. A positive feedback mechanism.



Key structural results:

Mean degree:

$\langle k \rangle = 2m \quad$ (edge-counting argument)

Power-law tail:

$P(k) \sim 2m^2 k^{-3} \quad (\tau = 3$, independent of $m)$

Hubs visible:

$n = 80$ BA graph shows nodes with degree 15–22 alongside most nodes at degree 2–4

Barabási–Albert: Scale-Free Networks

$$P(k) \sim k^{-3}$$

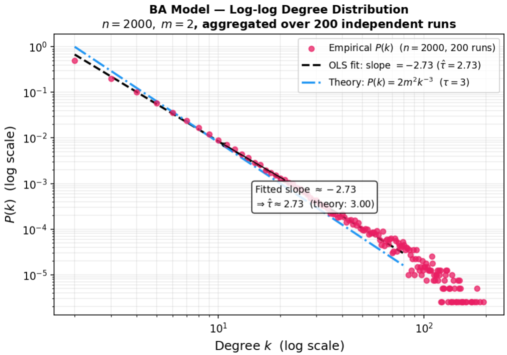
The "Power Law" Signature

- **Heavy Tail:** A few "hubs" hold the vast majority of connections.
- **Scale-Free:** No "typical" node degree (variance is infinite in the limit).
- **Resilience:** Robust to random failure, but fragile under targeted hub attacks.

Why it matters: Unlike ER or WS, this model explains the "hubs" we see on the World Wide Web and citation networks.

BA Simulation: Power-Law Degree Distribution (RQ3)

200 runs · $n = 2000$ · $m = 2$ · OLS fit over $k \in [2, 50]$ in log-log space



RQ3 Answer

-2.73
OLS fitted slope
($R^2 = 0.998$)

$\tau = 3$
Theory prediction
(asymptotic)

$\langle k \rangle = 2m$
Verified across
$m \in \{1, 2, 3, 5, 8, 12\}$

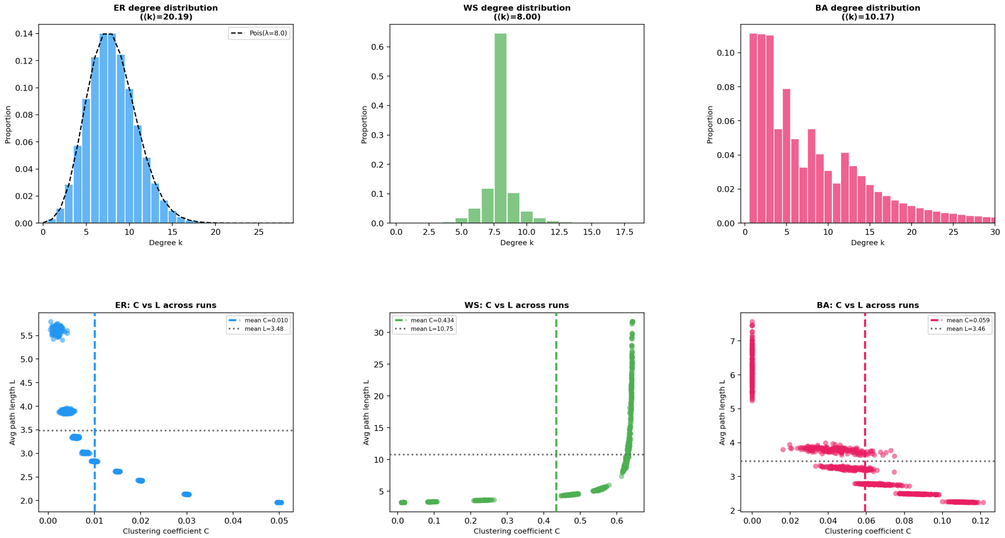
Slope of $-2.73 < -3$ due to finite-n tail truncation at $k \approx$ 50–80. As $n \to \infty$ slope converges to -3.

Comparative Structural Properties

	Erdős–Rényi	Watts–Strogatz	Barabási–Albert
Mechanism	Independent edge prob. p	Rewiring a ring lattice with prob. β	Growth + preferential attachment
Degree dist.	Binomial $\to$ Poisson(λ)	Near-symmetric, narrow spread	Power law $P(k) \sim k^{-3}$
Clustering	Low ($\approx p$)	High for small β	Low–moderate
Hubs	None	None	Present (heavy tail)
Path length	$O(\log n / \log \log n)$	$O(\log n)$ in small-world regime	Short due to hub routing
Mean degree	$\langle k \rangle = (n-1)p$	Exactly preserved at k	$\langle k \rangle = 2m$
Giant component	Threshold at $p_c = 1/n$	Always 1 (edges preserved)	Always 1 (by construction)
Typical uses	Thresholds, percolation theory	Social, neural networks	Web, citation, scale-free nets

Model Comparison: ER | WS | BA

Top: degree distributions   Bottom: clustering vs path length



Conclusions: Research Questions Answered

- **RQ1 (Phase Transition):** Threshold confirmed at $p \approx 0.0014$. Discrepancy from theory (0.001) is due to **finite-size effects**.
- **RQ2 (Small-World):** Short paths form at $\beta = 10^{-3}$ while local clustering remains high.
- **RQ3 (Scale-Free):** Power law slope measured at **-2.73**; converges to theoretical **-3** as $n \to \infty$.

Takeaway: No single model is perfect. Real networks require a blend of local clustering (WS) and global hubs (BA).